Motion of the Planets and their Interactions

EART22001 Environmental Modelling

In this information sheet we review the ideas behind the planetary-motion practical. The aim is to use a physical system that is familiar in outline, but which quickly becomes complicated enough that numerical modelling is useful.

1 Two-body and many-body motion

There are analytical solutions for the **two-body problem**. Newton's laws of motion for two bodies predict elliptical planetary orbits, as described observationally by Kepler.

Once the gravitational interactions of several planets are included, there is no general analytical solution to the n**-body problem**. We therefore use a computer to solve many coupled ordinary differential equations.

When interactions with other planets are included, the orbits develop a slow **precession**: the orientation of an elliptical orbit changes slightly with time. Planets such as Jupiter and Saturn exert small gravitational pulls on the other planets and perturb their orbits. Orbital precession is one of the Milankovitch cycles; changes in Earth's orbit on these long timescales are relevant to long-term changes in climate.

2 What we will do in the practical

We will run a Solar System model and plot the orbits of the planets as a function of time. We will then use the model output to introduce **Fourier analysis**.

Fourier analysis allows us to identify frequencies present in an oscillating time series. In planetary motion, the main frequency is associated with the orbital period: the time a planet takes to travel around the Sun. Interactions with other planets can introduce additional frequencies. Jupiter is particularly important because of its large mass and its approximately 11.8-year orbital period.

We will use these ideas to look for signatures of planetary interactions in the model output.

3 Planetary interactions and the Sun's wobble

The model allows us to examine effects that arise because the planets interact gravitationally rather than moving independently. One example discussed in the lecture is the **Great Conjunction**, when Jupiter and Saturn are aligned in the sky.

Lastly, we will run the model and analyse the small **wobble of the Sun** caused by the gravitational pull of the planets. Similar stellar wobbles can be used to identify exoplanets that are too small or faint to be observed directly: rather than seeing the planet itself, we can detect its influence on the motion of its parent star.

The modelling thread. In this exercise we move from Newton's laws, to a set of coupled ordinary differential equations, to numerical model output, and finally to time-series analysis. The practical is therefore both a planetary-motion example and an introduction to a modelling workflow that we will use again in the course.

4 Before the practical

Before the practical class:

- review this information sheet;
- make sure you know how to log in to the modelling server using a Terminal, PowerShell or Command Prompt window;
- complete the previous practical if you have not already done so.